



Kinetics of C_5H_4 isomer + H reactions and incorporation of C_5H_x ($x = 3 - 5$) chemistry into a detailed chemical kinetic model

Rasheed Adewale, Gabriel da Silva*

Department of Chemical Engineering, The University of Melbourne, Victoria 3010 Australia



ARTICLE INFO

Article history:

Received 16 June 2020

Revised 28 December 2020

Accepted 29 December 2020

Available online 16 January 2021

Keywords:

Potential energy surface

Detailed chemical kinetic models

Toluene flame

C_5H_x species

1-Vinylpropargyl

ABSTRACT

Although C_5H_4 isomers are detected in flames, they are not thoroughly incorporated into detailed chemical kinetics models (DCKMs). Here we use RRKM/ME modelling to simulate $C_5H_4 + H$ reactions on a C_5H_5 potential energy surface. Kinetic studies indicate that $C_3H_3 + C_2H_2$ is the main fate but fall-off from the initial adduct isomer back to $C_5H_4 + H$ cannot be ignored at relevant combustion temperatures of 900 to 2000 K. Calculated rate coefficient expressions were incorporated into a DCKM for a toluene flame, along with updates to other relevant reactions from the recent literature, particularly the open-chain 1-vinylpropargyl radical, $l-C_5H_5$. Obtained species mole fractions were found to be in good agreement with published experimental data for a low-pressure toluene flame, with a significant improvement in predicted concentration of the cyclopentadienyl radical. The presented DCKM will allow for further reactions of C_5H_x species such as 1-vinylpropargyl to be included in combustion simulations.

© 2021 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Polycyclic Aromatic Hydrocarbons (PAHs) and soot particles form considerable portions of incomplete combustion exhaust gas streams. PAHs are potentially tumorigenic and mutagenic [1–3] and have been linked to lung cancer [4]. Soot particles believed to be formed from PAHs, greatly impact air quality, and are promoters of global warming [5]. Due to these implications, significant efforts have gone into abating PAHs and soot particle emissions from combustion processes. Detailed chemical kinetic models (DCKMs) describing these processes are required to design more efficient combustion processes and formulate improved fuel blends to address the problems highlighted above. Despite substantial efforts, the elementary chemical steps responsible for the formation and growth of PAHs into particulate matter are yet to be unraveled.

C_5H_x ($x = 2 - 6$) species, especially C_5H_5 , C_5H_3 , and C_5H_4 , have been identified to play important roles as intermediates in molecular growth chemistry in hydrocarbon flames [6,7]. Their presence in different isomeric forms has been confirmed in fuel-rich flames of allene, propyne, cyclopentene, and benzene, using photoionization mass spectrometry [8]. C_5H_5 species, particularly the $c-C_5H_5$ isomer, cyclopentadienyl, are believed to be the most important C_5H_x species involved in hydrocarbon molecular growth to PAHs and then soot. It has been widely accepted that $c-C_5H_5$ radicals un-

dergo self-recombination to form naphthalene, and their reaction with acetylene is also a feasible pathway for benzyl radical formation in fuel-rich flames [8,9]. C_5H_3 species, similar to propargyl due to their resonance stability and strong C–H bonds [8], are believed to react slowly with CH_3 to form benzene, fulvene, or phenyl (+ H) in flames [10–12]. In a crossed molecular beam experimental study combined with *ab initio* calculations, Guo et al. [13] revealed that dicarbon molecules can react with allene to yield C_5H_3 isomers, $HCCCCCH_2$ and $HCCCHCCH$, via C_5H_4 intermediates $H_2CCCCCH_2$ (1,2,3,4-pentatetraene) and $HCCCH_2CCH$ (1,4-pentadiyne), respectively. This alludes to C_5H_4 isomers playing key roles as intermediates in flames and in interstellar environments by contributing to PAH growth, leading to soot and aerosol formation [14,15]. C_5H_4 species also serve as a source for methyl radicals when they react with ethynyl in combustion flames [14]. Therefore, it is vital to capture the interplay between C_5H_x species in combustion models to quantify their effects in flame chemistry. While there has been progress in the description of C_5H_5 species in DCKMs, C_5H_4 isomers have not been rigorously incorporated into combustion models, thereby hindering the development of their chemistry.

Recent work has expanded our knowledge of the C_5H_5 potential energy surface, upon which the chemically activated $C_5H_4 + H$ reactions would take place. An experimental investigation by Shapero et al. [16] reveals that two channels exist for the photodissociation of cyclopentadienyl ($c-C_5H_5$) upon excitation at 248 nm: $C_5H_4 + H$, where the C_5H_4 isomer was identified as ethynylallene, and $C_3H_3 + C_2H_2$. Previous studies on C_5H_5 decomposition have only considered $C_3H_3 + C_2H_2$ as the fragments [17–19]. More

* Corresponding author.

E-mail address: gdasilva@unimelb.edu.au (G. da Silva).

recently, an experimental investigation, by Savee et al. [20], on consecutive addition of C_2H_2 to C_3H_3 also observed the formation of an open chain $I-C_5H_5$ isomer (1-vinylpropargyl) alongside the traditionally reported $c-C_5H_5$ (cyclopentadienyl) product. There was evidence for acetylene addition to $I-C_5H_5$ leading to an unknown C_7H_7 isomer, which may be the vinylpropargyl chromophore incorporated within a cyclopentadienyl ring structure. The absence of $I-C_5H_5$ (1-vinylpropargyl) from combustion models precludes the incorporation of this new chemistry.

Other experimental investigations have also probed the role of C_5H_5 in bimolecular reactions involving $C_3H_3 + C_2H_2$, $C_2H + CH_3CCH$, and $C_2H + H_2CCCH_2$ [20–22]. Moskaleva and Lin [23] carried out a thorough theoretical study on C_5H_5 but C_3H_3 and C_2H_2 was the only pathway investigated while Jamal and Mebel [24] published a more comprehensive surface with additional C_5H_5 decomposition fragments, and da Silva recently included the isomerization mechanism between $c-C_5H_5$ and $I-C_5H_5$ into a theoretical kinetic study of C_5H_5 decomposition and the reverse $C_3H_3 + C_2H_2$ reaction. However, the $C_5H_4 + H$ pathway accessing both the cyclic and linear C_5H_5 isomer surfaces is yet to be incorporated into chemical kinetic investigations.

In extending the work of da Silva [25], the association reaction of $C_5H_4 + H$ to both cyclopentadienyl ($c-C_5H_5$) and (1-vinylpropargyl) ($I-C_5H_5$) is investigated using computational quantum chemistry techniques to understand their formation mechanism. Branching ratios of product species are calculated as a function of temperature and pressure representative of combustion systems using statistical reaction rate theory modelling. Kinetic rate parameters of the different species reactions involved are coupled into a chemical kinetic model, for toluene, published by Yuan et al. [26,27]. Kinetic rate data of other species important to C_5H_x chemistry were sourced from literature to allow for a rigorous description and improvement of the DCKM. Chemical kinetic simulations were performed using the resulting model to predict oxidation and PAH evolution profiles in a toluene flame, with predictions compared to available experimental data [26].

2. Methods

2.1. Ab initio calculations

Quantum chemistry computations were employed in Gaussian 16 [28]. The geometries, frequencies, and energies of all reactants, products, intermediates, and transition states were calculated and optimized using the G3X-K composite theoretical method [29] which is specifically designed for thermochemical kinetics and is capable of predicting barrier heights and enthalpies of formation with ~ 0.5 kcal mol $^{-1}$ accuracy. In this method, M06–2X/6–31G(2df,p) optimized structures are used in a series of single-point energy calculations starting from HF and terminates at the CCSD(T) theory with incrementally decreasing size basis sets. The G3X-K energy is then obtained from a prescribed combination of the empirically scaled zero-point energies. Optimized structures are checked for the absence of an imaginary frequency for the minima and transition states are confirmed to have only one imaginary frequency. Also, transition state connectivity was validated through intrinsic reaction coordinate scans.

2.2. Master equation calculations

Statistical reaction rate calculations were carried out using the Multiwell-2017 suite [30], to obtain temperature- and pressure-dependent phenomenological rate coefficient predictions in the $C_5H_4 + H$ reaction process. Calculations utilized the M06–2X/6–31G(2df,p) structural parameters and G3X-K energies. Internal degrees of freedom were modeled as harmonic oscillators, except

for low-energy internal rotations which were described as free rotors. For $c-C_5H_5$ (cyclopentadienyl) the Jahn-Teller effect was accounted for by using the scaled vibrational frequencies of Kiefer et al. [31], as described previously [32]. External rotation was treated using an active 1-D and an inactive 2-D rotor (the symmetric top assumption). Sums and densities of states were obtained from Stein-Rabinovitch-Beyer-Swinehart counts. Rice-Ramsperger-Kassel-Marcus (RRKM) theory was used to determine microcanonical rate coefficients, $k(E)$, with tunneling corrections based on the 1-D unsymmetrical Eckart barrier used in H-shift reactions. Collisional energy transfer between C_5H_5 isomers and the N_2 bath gas was modeled using the energy dependent biexponential model $\Delta E_d = 65 + 0.0058E$ cm $^{-1}$, which was experimentally determined for toluene in N_2 [33]. Lennard-Jones parameters for all C_5H_5 species were taken to be 5 Å for σ and 300 K for ϵ/k_b . These are based on values estimated using the ANSYS Chemkin-Pro 19.0 Reaction Workbench [34] property estimation functionality, which uses critical property correlations of Chung et al. [35]. The stochastic master equation calculations were performed using a grain size of 10 cm $^{-1}$ with a total of 2000 energy grains, after which the quasi-continuum component of the hybrid master equation formulation was carried up to 150 000 cm $^{-1}$. Simulation of the chemically activated $C_5H_4 + H$ reactions consisted of 10^6 trials and sufficient collisions to ensure that steady-state or the thermal decay regime [36] is reached. Phenomenological $k(T, P)$ values were extracted using the PPM code [36], ensuring that they correspond only to the chemically activated (and not thermally activated) process. Due to longer computational times, 10^5 independent trajectories were used for unimolecular reactions and rate coefficients were obtained after an initial relaxation period, ensuring that they corresponded to first order decay [37].

2.3. Chemical kinetic modelling

Chemical kinetic simulations were carried out using the previously published mechanism of Yuan et al. [26] as a base model. The base “Yuan” model consisted of reactions of C_8 to polyaromatic species, toluene, benzene, 1,3-cyclopentadiene, and C_0 – C_4 species, and was validated using experimental data [26,27] for a low-pressure toluene flame. This model was updated by introducing newly calculated rate coefficients from this work, along with literature parameters for several related reactions relevant to toluene combustion. A key feature of the updated model referred to as the “present” model, is the treatment of the chemically activated $C_5H_4 + H$ reactions on the C_5H_5 potential energy surface, along with other C_5H_x reactions. Additionally, because the C_7H_5 radical fulvenallenyl is a major source of $c-C_5H_5$, as well as molecular weight growth products [38], this chemistry has also been updated. Although the base model was primarily developed for low-pressure flames, we have attempted to incorporate pressure dependence where possible. Tables 1 and 2 show the list of additions and modifications made to the base model to obtain the new kinetic model of 289 species and 1717 reactions used in this study. The kinetic model, which is provided as supporting material, groups together the new additions into updated pressure dependent C_5H_x and C_7H_x sub-models for easy incorporation into other DCKMs.

Flame modeling was conducted using the Premixed Laminar Burner-Stabilized Flame application in Chemkin-Pro 19.0 [39]. The experimental conditions - pressure = 4 kPa, % toluene = 8.14, % air = 41.86, % argon = 0.5, inlet mixture flow velocity = 35 cm s $^{-1}$ and the temperature profile reported by Li et al. [40] for a premixed toluene flame with an equivalence ratio of 1.75, were used as input parameters. The Soret effect, which takes thermal diffusion into consideration, and mixed-averaged transport options were used in the simulation. Also, the numerical scheme for convective flux was employed to conserve species fluxes across

Table 1

Reactions added and modified in the base mechanism used in the present work in the form $k = AT^n \exp(-E_a/RT)$; AT^n in $\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ or s^{-1} , E_a in cal mol^{-1} and P in atm.

		P	A	n	E_a	Ref.
Added reactions						
R1	$\text{CHCCHCCH}_2 + \text{H} \leftrightarrow \text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$	0.01	2.26×10^{-9}	-0.771	6818.2	this work
		0.1	4.65×10^{-9}	-1.134	7793.32	
		1	6.25×10^{-10}	0	6813.5	
		10	1.45×10^{-10}	0.185	6975.3	
		100	2.58×10^{-9}	-0.259	11,100	
R2	$\text{CHCCHCCH}_2 + \text{H} \leftrightarrow \text{I-C}_5\text{H}_5$ (1-vinylpropargyl)	0.01	4.17×10^{-11}	-6.243	2541.1	this work
		0.1	5.25×10^{-12}	-2.568	1270.521	
		1	2.98×10^{-5}	-9.3	10,626	
		10	1.62×10^{37}	-14.708	20,702	
		100	1.99×10^{32}	-12.716	26,575	
R3	$\text{CHCCHCCH}_2 + \text{H} \leftrightarrow \text{c-C}_5\text{H}_5$ (cyclopentadienyl)	0.01	1.39×10^{-3}	-12.626	12,970.47	this work
		0.1	1.39×10^{-3}	-11.272	13,163.29	
		1	1.13×10^{-2}	-12.5	14,738.1	
		10	2.25×10^{-8}	-0.896	4135.7	
		100	3.52×10^{-9}	-0.597	8063.8	
R4	$\text{CHCCHCCH}_2 + \text{H} \leftrightarrow \text{CHCCH}_2\text{CCH} + \text{H}$	0.01	3.72×10^{-14}	2.177	9629.86	this work
		0.1	3.72×10^{-14}	2.204	10,006.17	
		1	1.32×10^{-11}	0	16,621.8	
		10	4.52×10^{-3}	-2.339	24,427	
		100	7.75×10^{26}	-10.316	56,493	
R5	$\text{CHCCH}_2\text{CCH} + \text{H} \leftrightarrow \text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$	0.01	5.90×10^{-12}	1.052	4351.72	this work
		0.1	4.99×10^{-12}	1.196	4520.0	
		1	5.20×10^{-12}	1.195	4646.24	
		10	5.59×10^{-16}	1.489	4241.3	
		100	1.51×10^{-14}	1.083	5187.7	
R6	$\text{CHCCH}_2\text{CCH} + \text{H} \leftrightarrow \text{I-C}_5\text{H}_5$	0.01	2.49×10^{-5}	-18.153	14,553.3	this work
		0.1	2.32×10^{-8}	-9.272	9891.2	
		1	1.09×10^{-5}	-11.1	13,391	
		10	1.29×10^{-4}	-2.590	5837.4	
		100	4.40×10^3	-4.736	9801.4	
R7	$\text{CHCCH}_2\text{CCH} + \text{H} \leftrightarrow \text{c-C}_5\text{H}_5$	0.01	2.04×10^{-8}	-11.621	9583.1	this work
		0.1	1.17×10^{-6}	-11.049	11,863.9	
		1	3.61×10^{-4}	-13.1	15,627.9	
		10	1.84×10^6	-5.468	11,959	
		100	2.10×10^{-12}	-0.00493	4407.5	
R8	$n\text{-C}_5\text{H}_3 + \text{H} \leftrightarrow \text{CHCCHCCH}_2$		3.64×10^{-11}	0.251	200.36	[42]
R9	$n\text{-C}_5\text{H}_3 + \text{H} \leftrightarrow \text{C}_5\text{H}_4$ (CHCCH ₂ CCH)		1.02×10^{-11}	0.309	250.27	[42]
R10	$\text{C}_7\text{H}_4 + \text{H} \leftrightarrow \text{C}_7\text{H}_5$		1.73×10^{11}	-6.84	8480	[36]
R11	$\text{C}_7\text{H}_4 + \text{H} \leftrightarrow \text{C}_3\text{H}_3 + \text{C}_4\text{H}_2$		3.11×10^{-15}	1.61	5750	[36]
R12	$\text{C}_7\text{H}_4 + \text{H} \leftrightarrow i\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2$		1.09×10^{-10}	0.0786	6840	[36]
R13	$\text{C}_3\text{H}_3 + \text{C}_4\text{H}_2 \leftrightarrow \text{CHCCCCCHCCH}_2 + \text{H}$	0.001	3.79×10^{-6}	-1.685	27,800	[43]
		0.01	5.31×10^{-6}	-1.695	28,380	
		0.1	1.23×10^{-5}	-1.843	27,950	
		1	2.92×10^{-3}	-2.417	31,630	
		10	1.13×10^6	-4.817	38,300	
		100	1.18×10^0	-3.004	38,350	
R14	$\text{C}_3\text{H}_3 + \text{C}_4\text{H}_2 \leftrightarrow n\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2$	0.001	2.08×10^8	-6.022	23,020	[43]
		0.01	1.49×10^{19}	-9.112	29,920	
		0.1	2.97×10^{21}	-9.766	32,540	
		1	5.41×10^{28}	-11.552	42,900	
		10	2.97×10^{50}	-17.388	65,800	
		100	4.30×10^{75}	-23.998	97,400	
R15	$\text{C}_3\text{H}_3 + \text{C}_4\text{H}_2 \leftrightarrow i\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2$	0.001	5.76×10^{11}	-6.965	21,620	[43]
		0.01	1.59×10^{14}	-7.602	24,520	
		0.1	2.16×10^{26}	-11.164	31,860	
		1	5.91×10^{31}	-12.345	42,350	
		10	8.15×10^{37}	-13.8	54,510	
		100	2.77×10^{47}	-16.106	73,350	
R16	$i\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2 \leftrightarrow n\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2$	0.001	6.66×10^2	-4.543	23,510	[43]
		0.01	1.74×10^7	-5.768	27,020	
		0.1	7.64×10^{22}	-10.199	37,520	
		1	8.19×10^{30}	-12.3	46,880	
		10	1.38×10^{28}	-11.231	52,370	
		100	1.38×10^{48}	-16.509	76,690	
R17	$i\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2 \leftrightarrow \text{CH}_2\text{CCCCCHCCH} + \text{H}$	0.001	1.12×10^{-6}	-1.645	35,450	[43]
		0.01	3.52×10^{-5}	-2.042	36,890	
		0.1	4.02×10^{-2}	-2.95	37,950	
		1	4.72×10^{-1}	-3.266	38,670	
		10	2.81×10^5	-4.943	42,020	
		100	4.53×10^7	-5.27	46,430	
R18	$i\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2 \leftrightarrow \text{CH}_2\text{CC}(\text{CCH})\text{CCH} + \text{H}$	0.001	4.53×10^7	-5.27	46,430	[39]
		0.01	1.31×10^{13}	-6.919	47,870	

(continued on next page)

Table 1 (continued)

		<i>P</i>	<i>A</i>	<i>n</i>	<i>E_a</i>	Ref.
		0.1	3.50×10^{15}	−7.655	48,740	
		1	1.26×10^{22}	−9.424	54,740	
		10	2.31×10^{24}	−10.15	54,930	
		100	1.42×10^{22}	−9.448	55,200	
R19	$n\text{-C}_5\text{H}_3 + \text{C}_2\text{H}_2 \leftrightarrow \text{CH}(\text{CCH})_3 + \text{H}$	0.001	4.33×10^{20}	−8.916	65,610	[39]
		0.01	3.51×10^{21}	−9.103	68,110	
		0.1	1.33×10^{22}	−9.305	67,470	
		1	1.88×10^{23}	−9.656	67,490	
		10	6.93×10^{23}	−9.569	74,240	
		100	1.07×10^{23}	−9.466	62,750	
R20	$\text{C}_2\text{H}_2 + c\text{-C}_5\text{H}_5 \rightarrow \text{A1CH}_2 \text{ (benzyl)}$		1.40×10^{17}	−1.300	31,000	[44]
	$\text{A1CH}_2 \rightarrow \text{C}_2\text{H}_2 + c\text{-C}_5\text{H}_5$		4.80×10^{12}	0.00	72,500	
R21	$c\text{-C}_5\text{H}_5 + \text{OH} \rightarrow \text{CO} + \text{C}_2\text{H}_2 + \text{C}_2\text{H}_4$		1.00×10^{14}	0.250	4350.0	[45]
R22	$c\text{-C}_5\text{H}_5 + \text{OH} \rightarrow \text{H} + \text{CH}_2\text{CO} + \text{C}_3\text{H}_3$		3.50×10^{13}	0.250	4350.0	[45]
R23	$c\text{-C}_5\text{H}_5 + \text{OH} \rightarrow \text{C}_5\text{H}_5\text{O} \text{ (1,3-Cyclopentadiene-1-yloxy)} + \text{H}$	0.0395	2.55×10^{31}	−4.98	16,090	[46]
		1	3.48×10^{25}	−3.18	17,320	
		10	6.73×10^{13}	0.20	13,690	
		100	3.31×10	3.69	9420	
R24	$c\text{-C}_5\text{H}_5 + \text{OH} \rightarrow \text{C}_5\text{H}_5\text{O} \text{ (1,4-Cyclopentadiene-2-yloxy)} + \text{H}$	0.0395	9.47×10^{25}	−3.56	17,490	[46]
		1	6.37×10^{23}	−2.80	20,740	
		10	4.12×10^{14}	−0.11	18,980	
		100	3.60×10^4	2.80	17,500	
R25	$c\text{-C}_5\text{H}_5 + \text{OH} \rightarrow \text{C}_5\text{H}_4\text{OH} + \text{H}$	0.0395	1.75×10^{26}	−3.49	16,630	[46]
		1	1.76×10^{23}	−2.49	19,380	
		10	1.06×10^{12}	0.76	15,960	
		100	6.56×10^{-4}	5.05	9260	
R26	$c\text{-C}_5\text{H}_5 + \text{O}_2 \rightarrow \text{HCO} + \text{C}_4\text{H}_4\text{O}$	0.03	6.87×10^8	0.61	23,783	[47]
		1	1.11×10^{12}	−0.28	26,565	
		10	2.62×10^{13}	−0.65	27,939	
		100	1.99×10^{25}	−3.79	41,788	
Modified reactions						
R27	$c\text{-C}_5\text{H}_5 + \text{H} \leftrightarrow \text{C}_5\text{H}_6$		2.71×10^{63}	−14.79	21,050	[48]
R28	$c\text{-C}_5\text{H}_5 + \text{O}_2 \rightarrow \text{C}_5\text{H}_4\text{O} + \text{OH}$		8.77×10^{01}	3.11	23,496	[47]
R29	$c\text{-C}_5\text{H}_5 + \text{O} \rightarrow \text{CO} + n\text{C}_4\text{H}_5$	0.0395	4.85×10^{14}	−0.26	5560	[49]
		1	4.87×10^{14}	−0.26	5570	
		10	5.75×10^{14}	−0.28	6180	
		100	8.43×10^{14}	−0.32	8670	
R30	$c\text{-C}_5\text{H}_5 + \text{OH} \rightarrow \text{CO} + \text{C}_4\text{H}_6$	0.0395	2.69×10^{34}	−6.07	15,340	[46]
		1	7.55×10^{27}	−4.07	16,610	
		10	1.16×10^{20}	−1.75	16,780	
		100	1.59×10^8	1.68	15,560	
R31	$c\text{-C}_5\text{H}_5 + \text{O}_2 \rightarrow \text{C}_5\text{H}_5\text{O} \text{ (2,4-Cyclopentadiene-2-yloxy)} + \text{H}$	0.03	1.06×10^{14}	−0.07	50,911	[47]
		1	1.06×10^{15}	−0.34	51,792	
		10	1.06×10^{15}	−0.34	51,792	
		100	1.06×10^{15}	−0.34	51,792	
R32	$c\text{-C}_5\text{H}_5 \leftrightarrow l\text{-C}_5\text{H}_5$		1.84×10^{94}	−24.4	79,300	[25]
R33	$l\text{-C}_5\text{H}_5 \leftrightarrow \text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$		1.01×10^{68}	−16.1	83,000	[25]
R34	$\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2 \rightarrow c\text{-C}_5\text{H}_5$		7.55×10^{99}	−24.4	128,200	[25]
	$c\text{-C}_5\text{H}_5 \rightarrow \text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$		2.00×10^{12}	0.00	68,000	[45]
R35	$i\text{-C}_4\text{H}_5 + \text{H} \leftrightarrow \text{C}_4\text{H}_6\text{-12 (1,2-butadiene)}$		8.28×10^{-11}	0.119	66.74	[38]
R36	$i\text{-C}_4\text{H}_5 + \text{H} \leftrightarrow \text{C}_4\text{H}_6 \text{ (1,3-butadiene)}$		5.30×10^{-12}	0.359	204.6	[38]
R37	$i\text{-C}_5\text{H}_3 + \text{H} \leftrightarrow \text{C}_5\text{H}_4\text{-13 (1,3-pentadiyne)}$		6.21×10^{-11}	0.145	115.4	[38]
R383	$i\text{-C}_5\text{H}_3 + \text{H} \leftrightarrow \text{C}_5\text{H}_4 \text{ (1,2,3,4-pentane tetraene)}$		8.29×10^{-12}	0.401	195.25	[38]
R39	$i\text{-C}_5\text{H}_3 + \text{H} \leftrightarrow \text{HCCHCCH}_2 \text{ (ethynylallene)}$		3.88×10^{-13}	0.681	450.88	[38]
R40	$\text{A1CH}_2 + \text{O} \leftrightarrow \text{A1CHO} + \text{H}$		4.566×10^{-11}	0.0723	732	[41]
R41	$\text{A1CH}_2 + \text{O} \leftrightarrow \text{A1} + \text{HCO}$		1.159×10^{-11}	0.329	686	[41]
R42	$\text{A1CH}_2 + \text{O} \leftrightarrow \text{OC}_6\text{H}_4\text{CH}_2 + \text{H}$		4.218×10^2	−3.72	13,780	[41]

the flame zone. To ensure that the simulation results are grid size-independent, a maximum grid point of 1000 was applied. The solution curvature and gradient, for the adaptive grid control, were both set to 0.9 for an initial run. The obtained solution from the initial run was then fed into a continuation run of the adaptive grid controls based on solution curvature and solution gradient in the pairs (0.7,0.7), (0.5,0.5), (0.5,0.4), (0.5,0.3), (0.5,0.2) and (0.5,0.1) respectively. This was done to enhance the convergence of the numerical solution.

The new kinetic model was also validated against experimental data, for toluene pyrolysis in a flow reactor, reported by Yuan et al. [27]. A detailed description of the implementation of the flow reactor simulation and modeling in Chemkin-Pro-software is provided in another work of the authors [41]. The model was validated over

a wide range of five experimental pressure – 5, 30, 80, 200, and 760 torr and the measured temperature profiles at the respective pressures. The model validation figures, supplementary Figs. 1–5, are provided in the supplementary materials and are not discussed here.

3. Results and discussion

3.1. Potential energy diagram

The potential energy diagram, shown in Fig. 1, delineates the mechanistic pathways for the $\text{C}_5\text{H}_4 + \text{H}$ reaction, proceeding via initial formation of the acetylenic and allenic adducts **1** and **7**, respectively. The potential energy surface is adapted from that

Table 2

Added and modified reactions updated in the base mechanism used in the present work in Troe [42] form $k[s^{-1}] = \{k^\infty/[1 + k^\infty/(k^0[M])]\}F_c$; $F_c = (1 - a)e^{(-T/b)} + ae^{(-T/c)} + e^{(-d/T)}$; $k = AT^n \exp(-E_a/RT)$; AT^n in $\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ or s^{-1} and E_a in cal mol^{-1} .

		A	n	E_a		Ref.
Added reactions						
R43	$\text{C}_7\text{H}_6 (+\text{M}) \leftrightarrow \text{C}_7\text{H}_5 + \text{H} (+\text{M})$					
	High-pressure limit (k^∞)	2.00×10^{16}	0	80,690		[34]
	Low-pressure limit (k^0)	1.95×10^5	-3.644	44,170		
	Troe parameters (a, b, c, d)	3.90×10^{-1}	6.60×10^1	3.22×10^3	6.81×10^4	
R44	$\text{C}_7\text{H}_5 (+\text{M}) \leftrightarrow \text{n-C}_5\text{H}_3 + \text{C}_2\text{H}_2 (+\text{M})$					
	High-pressure limit (k^∞)	1.12×10^{40}	-6.905	110,740		[34]
	Low-pressure limit (k^0)	8.9×10^{24}	-9.17	77,640		
	Troe parameters (a, b, c, d)	2.01×10^0	8.19×10^2	9.36×10^2	2.73×10^3	
Modified reactions						
R45	$\text{C}_7\text{H}_5 (+\text{M}) \leftrightarrow \text{C}_3\text{H}_3 + \text{C}_4\text{H}_2 (+\text{M})$					
	High-pressure limit (k^∞)	9.81×10^{39}	-6.93	98,050		[34]
	Low-pressure limit (k^0)	7.01×10^{24}	-9.163	58,130		
	Troe parameters (a, b, c, d)	7.49×10^0	4.37×10^2	4.16×10^2	2.43×10^3	
R46	$\text{C}_7\text{H}_5 (+\text{M}) \leftrightarrow \text{i-C}_5\text{H}_3 + \text{C}_2\text{H}_2 (+\text{M})$					
	High-pressure limit (k^∞)	8.47×10^{39}	-6.816	107,120		[34]
	Low-pressure limit (k^0)	6.17×10^{24}	-9.121	69,870		
	Troe parameters	2.00×10^0	8.46×10^2	8.77×10^2	2.88×10^3	

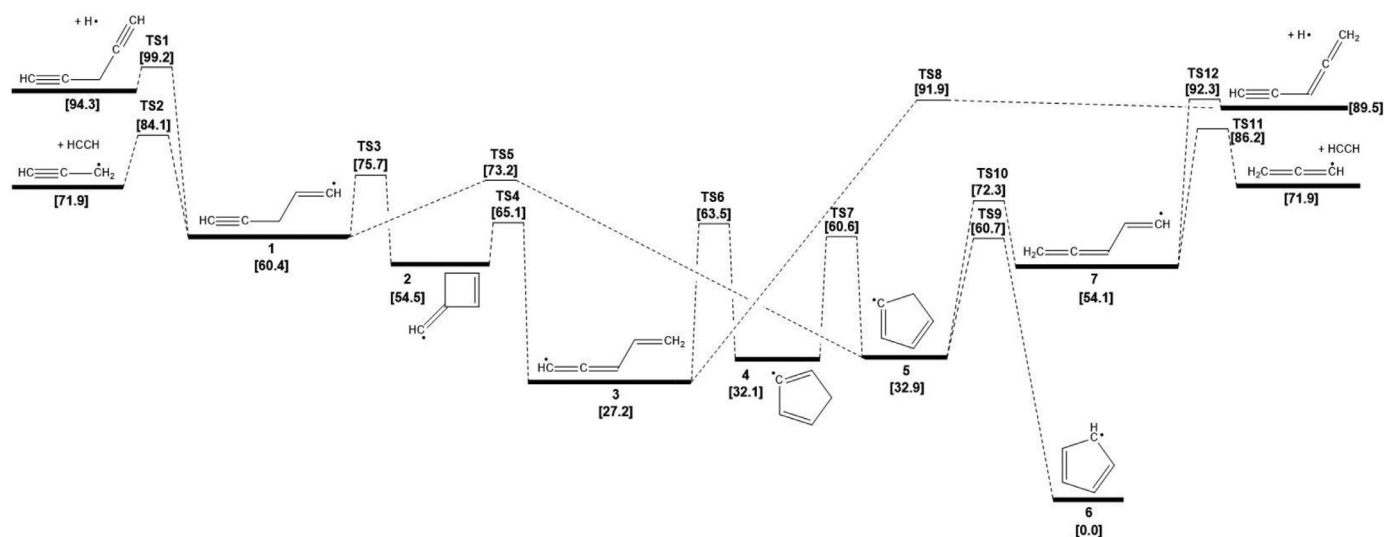


Fig. 1. C_5H_5 potential energy surface. Energies are 0 K enthalpies in kcal mol^{-1} , calculated at the G3X-K level of theory.

used by da Silva to model the $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ reaction [24], with the addition of the dissociation pathways to $\text{C}_5\text{H}_4 + \text{H}$. Note that the addition of the $\text{C}_5\text{H}_4 + \text{H}$ entry channels to the C_5H_5 potential energy diagram is not expected to affect the kinetic parameters calculated by da Silva [25]. This is because there is a significant barrier height of about 15 kcal/mol that needs to be overcome to produce $\text{C}_5\text{H}_4 + \text{H}$ fragments when the C_5H_5 energy surface is accessed from the $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ entry channel.

The addition of H to the inner C atom of the triple bond in the C_5H_4 isomer HCCH_2CCH (1,4-pentadiyne) to give the chemically activated radical adduct **1** proceeds, through **TS1**, with a barrier of about 5.0 kcal mol^{-1} . The C_5H_5 isomer **1** sits at ~34 kcal mol^{-1} below the initial reactants ($\text{HCCH}_2\text{CCH} + \text{H}$). In a subsequent isomerization step, through one of the two pathways available to it, **1** can ring-close to a five-membered ring isomer **5** in a step controlled by **TS5**, which sits at 26 kcal mol^{-1} and 21.1 kcal mol^{-1} below **TS1** and the initial reactant energies, respectively. Isomer **5** then undergoes a 1,2 H-migration to give **6** (*c*- C_5H_5), via **TS8**, with an activation energy of 38.5 kcal mol^{-1} below the entrance transition state; this represents the global minimum on the C_5H_5 potential energy surface. Alternatively, in a competitive reaction pathway, **1** can transform to a four-membered ring isomer **2**

via **TS3**, which resides at 13.5 kcal mol^{-1} below **TS1**. Subsequently, in a highly exothermic process, **2** overcomes a ring-opening barrier of 10.6 kcal mol^{-1} at **TS4** located 34.1 kcal mol^{-1} below the entrance transition state to produce *l*- C_5H_5 (**3**) in a locally deep well. *l*- C_5H_5 can later undergo a ring closure to form a five-membered ring compound **4** after surmounting a barrier of 36.3 kcal mol^{-1} . Isomer **4** can then transition into **5** through **TS7** in a 1,2 H-shift process before ring-opening to **7** (allenic adduct). Besides isomer **3**, that can also be formed via **TS10** at a barrier of 7.3 kcal mol^{-1} below **TS1**, **7** is also an entry channel to the C_5H_5 PES from the addition of H atom to another C_5H_4 isomer HCCCHCCH_2 (ethynylallene) via **TS12** with a barrier of 6.9 kcal mol^{-1} below **TS1**.

3.2. Rate coefficients

To determine the $\text{C}_5\text{H}_4 + \text{H}$ reaction kinetics, statistical reaction rate modelling was carried out, from 300 to 2000 K at 1 atm. Predicted branching fractions are shown in Fig. 2, with rate coefficients plotted in Fig. 3. Arrhenius fits to the rate coefficients are also depicted in Fig. 3, with the parameters listed in Table 1.

For $\text{HCCCHCCH}_2 + \text{H}$, the reaction occurs via initial formation of the chemically activated isomers **3** and **7**. From Fig. 2 (top plot), at

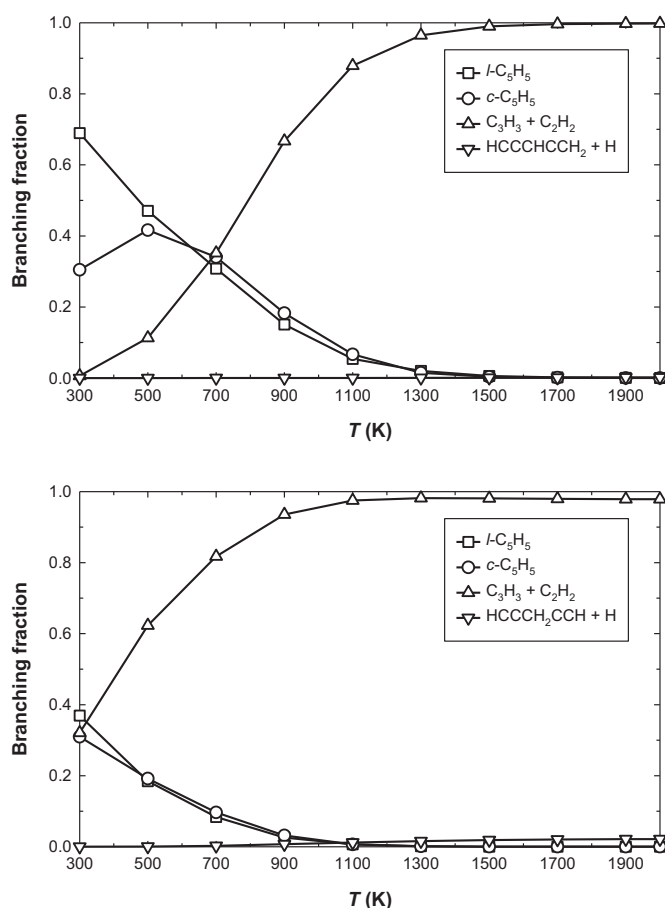


Fig. 2. Calculated branching fractions as a function of temperature in the CHCCH₂CCH + H reaction (top) and the HCCCHCCH₂ + H reaction (bottom) at 1 atm.

300 K, it is apparent that $l\text{-C}_5\text{H}_5$ is the dominant isomer at about 69% yield compared to ~31% for $c\text{-C}_5\text{H}_5$ with no production of $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ nor fall-off to the initial reactants HCCCHCCH₂ + H. Between 300 and 500 K, there is a steady increase in the yield of $c\text{-C}_5\text{H}_5$ and $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ with a noticeable decline in $l\text{-C}_5\text{H}_5$ yield. This reflects the decreasing ability to stabilize the relatively fragile open-chain species $l\text{-C}_5\text{H}_5$ with increasing energy, which further reacts to either become stabilized as the global minima cyclopentadienyl ($c\text{-C}_5\text{H}_5$) or dissociate to propargyl and acetylene. Post 500 K, the yield of $c\text{-C}_5\text{H}_5$ also begins to decline and from 700 K onwards, the dominant fate for the activated C_5H_5 population is for dissociation. Note that even at high temperatures the reverse dissociation to HCCCHCCH₂ + H (or to the other C_5H_4 isomer) is negligible, which can be attributed to the strong preference for the C_5H_5 adducts **3** and **7** to decompose to $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ over H atom loss.

The CHCCH₂CCH + H reaction evolves through the initial formation of isomer **1**. Qualitatively the behavior is identical to that of the HCCCHCCH₂ + H reaction, with $l\text{-C}_5\text{H}_5$ and $c\text{-C}_5\text{H}_5$ being stabilized at the lower temperature range and dissociating to $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ at higher temperatures. Compared to the HCCCHCCH₂ + H branching ratio profile, there is a significant formation of $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ even at 300 K, due to higher energy of the initiating C_5H_4 isomer. The fractional yield of $l\text{-C}_5\text{H}_5$ in this process seems to be more, albeit marginally, than that of $c\text{-C}_5\text{H}_5$ within the temperature range of 300 to 500 K. Subsequently, the two profiles ($l\text{-C}_5\text{H}_5$ and $c\text{-C}_5\text{H}_5$) reach about the same value at 500 K and then decline at similar rates to ~0% yield at 1100 K. This temperature

marks the onset of the fall-off back to CHCCH₂CCH + H with about 2% yield recorded between 1100 K to 2000 K. Within the temperature range of 300 to 1300 K, there is a reverse decomposition back to $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ which stabilizes at about 98% yield and remains at this value from ~1300 K to the maximum temperature of 2000 K.

Fig. 3, top plot, supports the branching ratio trends seen in Fig. 2 (top plot). The kinetics of the isomerization of adducts **3** and **7** to C_5H_5 isomers, $l\text{-C}_5\text{H}_5$ and $c\text{-C}_5\text{H}_5$, dominates the chemical process taking place between 300 K to 700 K after which the decomposition process to $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ overtakes and sustains its dominance throughout the moderate- to high-temperature regimes. It is also within this temperature domain that the kinetics of the fall-off back to HCCCHCCH₂ + H becomes relevant. From the onset (Fig. 3, bottom plot), within the temperature range considered in the CHCCH₂CCH + H mechanism, the decomposition of the addition adduct **1** to $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ appears to be the most significant chemical pathway. In contrast to the HCCCHCCH₂ + H system, the fall-off of the adducts back to the initial reactants seems to set-in earlier in the CHCCH₂CCH + H system. In both mechanisms (CHCCH₂CCH + H and HCCCHCCH₂ + H), the kinetics of the isomerization reaction to C_5H_5 isomers, $l\text{-C}_5\text{H}_5$ and $c\text{-C}_5\text{H}_5$, comes to a peak at around 700 K and begins to slow down afterward. This suggests that the decomposition of the C_5H_5 isomers, to acetylene and propargyl and C_5H_4 isomers (especially ethynylallene) + H, could be favorable channels at temperatures relevant to combustion.

3.3. Flame simulation

The flame simulations were compared and validated against published experimental data [40] for a premixed low-pressure toluene/ O_2 /Ar flame. As displayed in Fig. 4, Both our model and the Yuan model demonstrate very good performance in predicting the mole fractions of major reactants (C_8H_7 and O_2) and product (CO , CO_2 , H_2 , H_2O) species.

The primary decomposition or consumption of toluene follows several reaction pathways initiated, mainly, by H, O, and OH radicals to form phenyl, benzene, benzyl, methyl phenoxy, methylphenyl, etc., as primary intermediates. Based on the respective rate of production analyses obtained by integrating the rate of production of species between 0 - 12 mm, from the burner face, the two models gave quite similar toluene decomposition percentages to the respective primary intermediates. More than 76% of toluene produces benzyl via H-abstraction, with H atom attack contributing the most to toluene depletion with 53% (Yuan) and 52% (present), followed by O attack with 9% (Yuan) and 10% (present), and OH attack with 11% in each model. Both models predicted the production pathway of benzene from toluene, through H addition/methyl elimination, to be about 10% each.

As seen in Fig. 5, the experimental measurements and model predictions for two of the above-mentioned primary intermediates, benzyl, and benzene, together with four subsequent decomposition product species are compared.

The experimental mole fraction data trends are well captured by both models and are within the experimental measurement [40] uncertainty of $\pm 25\%$ for C_2H_2 (acetylene), C_4H_4 (vinylacetylene), and C_6H_6 (benzene), and $\pm 200\%$ for $\text{C}_6\text{H}_5\text{CH}_2$ (benzyl), $c\text{-C}_5\text{H}_5$ (cyclopentadienyl), and C_3H_3 (propargyl). Several species, mainly phenoxy, benzyl, phenyl, cyclopentadiene, etc., react in the radical pool to contribute to cyclopentadienyl radical production in the toluene flame [40]. This needs to be balanced with $c\text{-C}_5\text{H}_5$ consumption pathways in DCKMs to ensure satisfactory predictions. As seen, the Yuan model overpredicts the $c\text{-C}_5\text{H}_5$ mole fraction profile between 5 and 9 mm away from the burner while our model gives an excellent prediction of the experimental data. ROP analyses indicate that the two models give quite

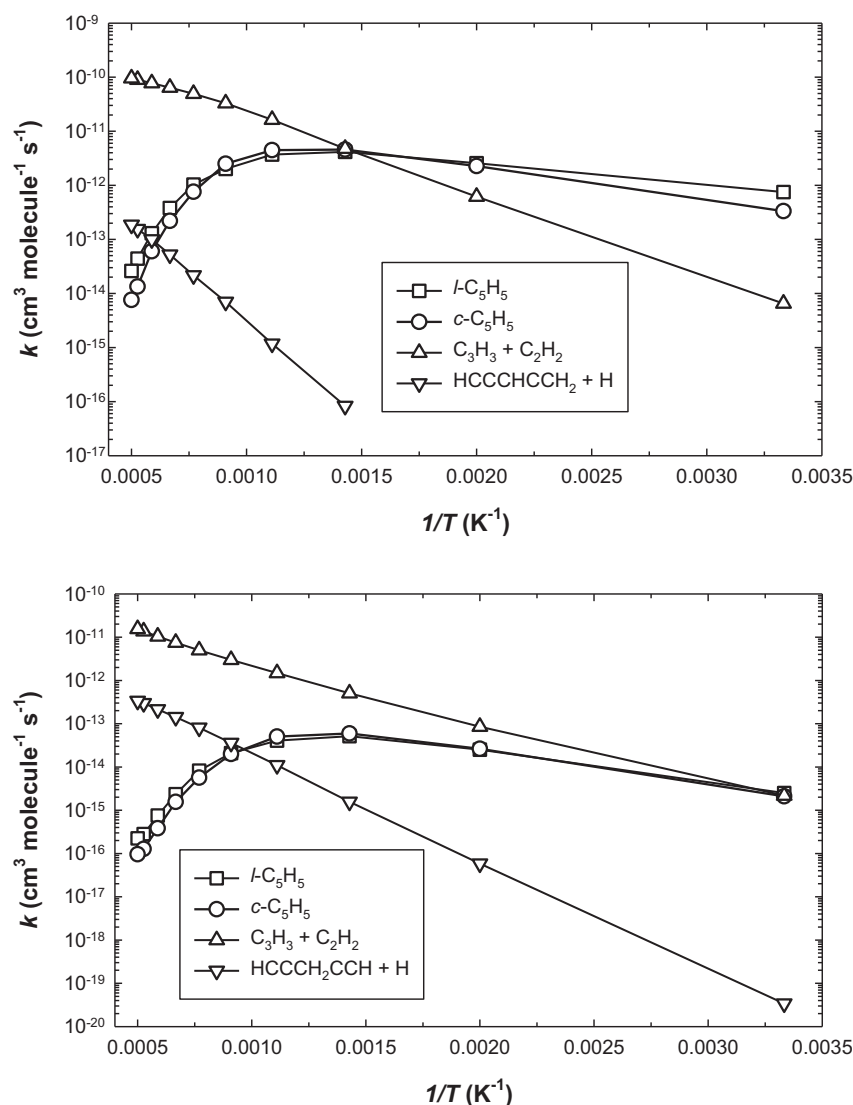


Fig. 3. Calculated rate coefficients (dot points) and model fits (solid lines) for the HCCCCHCH₂ + H reaction (top) and the CHCCH₂CCH + H reaction (bottom) at 1 atm.

similar percentage contributions as regards to *c*-C₅H₅ formation pathways, the most significant being through the decomposition of phenoxyl with 49% and 43% contribution in the Yuan model and the present study, respectively. This is followed by the reaction of cyclopentadiene with H (C₅H₆ + H ⇌ *c*-C₅H₅ + H₂) with 23% and 24% contribution towards *c*-C₅H₅ as predicted by the Yuan and the updated models, respectively. Considering the pathways leading to *c*-C₅H₅ consumption in the flame, the ROP analysis, with the new model, indicates that more than 11% of *c*-C₅H₅ isomerizes to its linear isomer, 1-vinylpropargyl (*l*-C₅H₅). A previous work [25] suggests that, at temperatures below the decomposition threshold, *c*-C₅H₅ may co-exist with its linear isomer-1-vinylpropargyl. Also, for the Yuan model, more than 90% of *c*-C₅H₅ decomposes to C₃H₃ and C₂H₂ compared to 55% with our model. In the updated model, cyclopentadienyl oxidation by OH attack, reactions R21 (*c*-C₅H₅ + OH → CO + C₂H₂ + C₂H₄) and R22 (*c*-C₅H₅ + OH → H + CH₂CO + C₃H₃) which are absent in the Yuan model contribute 35% and 12% respectively to *c*-C₅H₅ consumption. These discrepancies explain the lower *c*-C₅H₅ levels predicted by our model and are attributable to our improved description of *c*-C₅H₅ decomposition kinetics.

Aside from the *c*-C₅H₅, predicted mole fraction profiles of other species, on the C₅H₅ PES, were also compared to their experimen-

tal mole fraction profiles. Based on evaluated ROP data, the Yuan model predicts that more than 43% and 32% of C₃H₃ is generated from unimolecular *c*-C₅H₅ decomposition and the C₆H₅ + H reaction respectively, whereas our model estimated about 6% and 40% for the same channels. With the updated model, almost 7% of produced C₃H₃ and 3% of generated C₂H₂ originate from the unimolecular decomposition of 1-vinylpropargyl (*l*-C₅H₅) while its contribution, using the Yuan model, is negligible. The newly included CHCCH₂CCH + H reaction (Reaction R5), contributed about 4% and 2% to C₃H₃ and C₂H₂ production, respectively. This reaction is absent in the Yuan model or any other mechanism known to the authors. Predictions of the main consumption pathways for propargyl and acetylene in both models are almost comparable. In the Yuan and the present models, ca. 81% and 82%, respectively, of propargyl, is consumed through reaction with H/OH, to triplet propargylene (C₃H₂). Similarly, C₂H₂ consumption is mainly through its oxidation to ketylenyl (HCCO) with both models estimating a consumption percentage of ~ 69% each. Reactions R21 and R22 discussed above, give about 9% and 8% to the C₂H₂ and C₃H₃ pool, respectively.

In Fig. 6, the mole fraction profiles of large aromatics from C₈–C₁₄ are summarized with experimental measurement [40] uncertainties of a factor of 2 for phenanthrene (C₁₄H₁₀), naphtha-

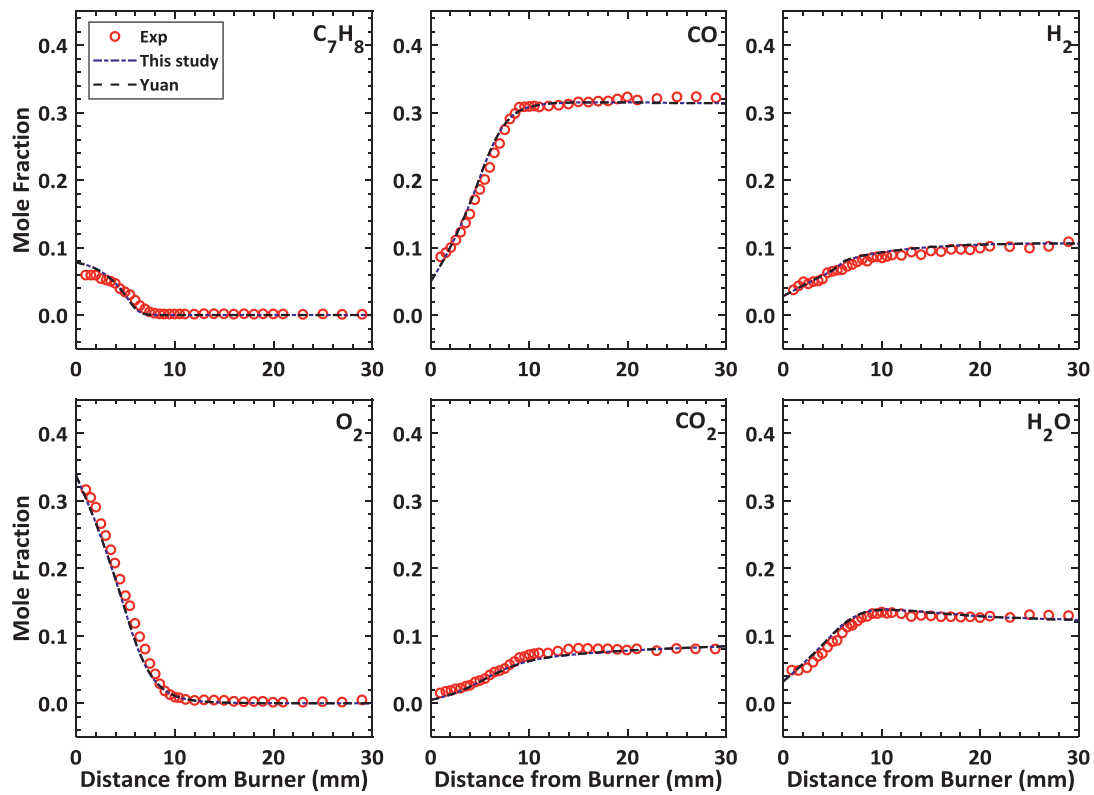


Fig. 4. Experimental (symbols) and simulated (dashed and dash-dotted lines for the Yuan and present models respectively) of major species in a premixed flame of toluene at an equivalence ratio of 1.75.

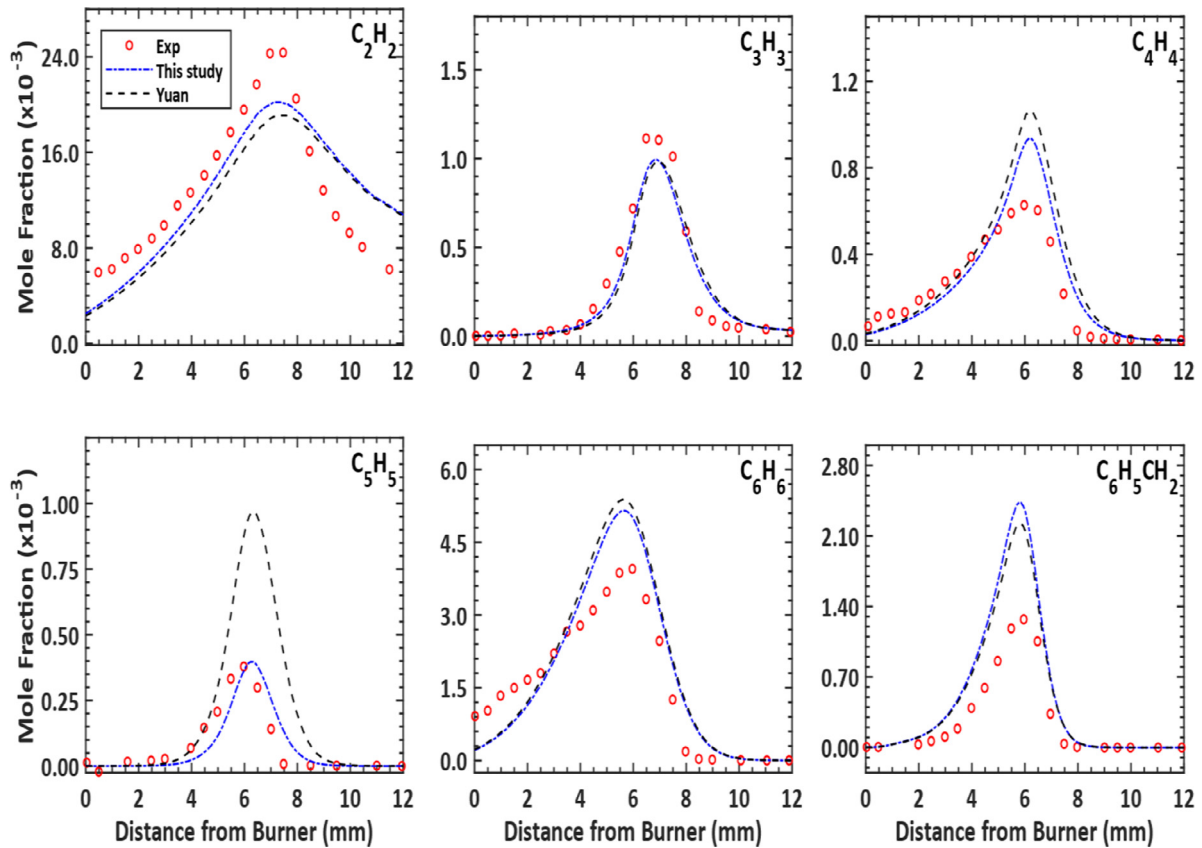


Fig. 5. Experimental (symbols) and simulated (dashed and dash-dotted lines for Yuan and present models, respectively) of C_2H_2 (acetylene), C_3H_3 (propargyl), C_4H_4 (vinylacetylene), C_5H_5 (c-cyclopentadienyl), C_6H_6 (benzene), and $C_6H_5CH_2$ (benzyl) in a premixed flame of toluene at an equivalence ratio of 1.75.

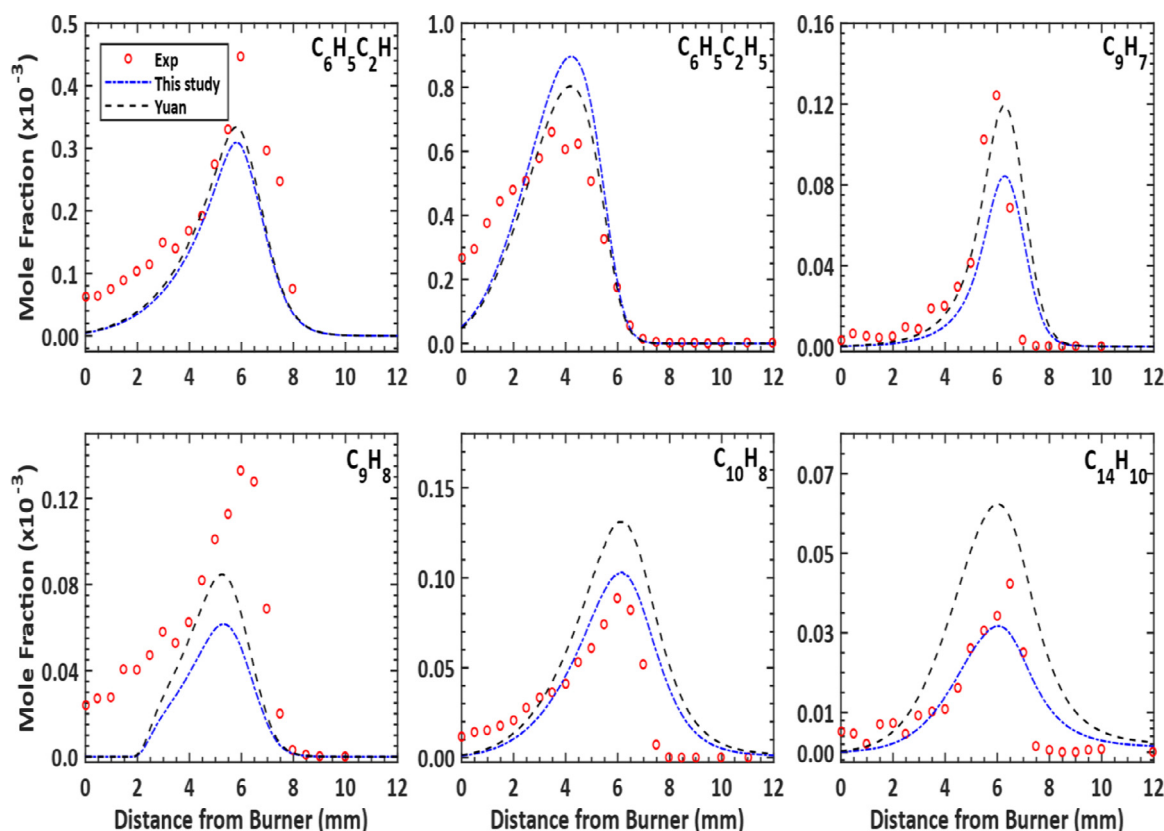


Fig. 6. Experimental (red symbols) and simulated (black dashed and blue dash-dotted lines for the Yuan and present models respectively) of $C_6H_5C_2H$ (phenylacetylene), $C_6H_5C_2H_5$ (ethylbenzene), C_9H_7 (indenyl), C_9H_8 (indene), $C_{10}H_8$ (naphthalene), and $C_{14}H_{10}$ (phenanthrene) in a premixed flame of toluene at an equivalence ratio of 1.75. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

lene ($C_{10}H_8$) and indenyl (C_9H_7), and $\pm 25\%$ for phenylacetylene, ethylbenzene, and indene. Both models give very good fits to the global trend of the experimental data within the reported error margins. The predicted mole fraction profiles, with both models, overlap for phenylacetylene, ethylbenzene, indenyl, and indene except within the intervals $\sim 2 - 6$ mm, $\sim 4 - 7$ mm, and $\sim 2 - 7$ mm from the burner face, respectively. With both the updated and Yuan models, the main production pathway of ethylbenzene, with 92% each, is through the reaction of methyl with benzyl while its major consumption pathway, with $\sim 54\%$ and 53% contribution with both the updated and Yuan models respectively, is through a two-step consecutive H loss to give styrene which is a precursor to phenylacetylene formation via a further two-step H loss reaction where $C_6H_5CCH_2$ and C_6H_5CHCH are intermediates.

For the two models, the most significant pathway for indenyl formation is through phenyl reaction with propargyl followed by H atom loss, with 56% and 45% contribution for the Yuan model and our model, respectively. Indenone formation through OH attack on indenyl is the most dominant consumption pathway while O attack, to produce C_6H_5CHCH and CO, is the second most significant route with 33% and 21% respectively as predicted by the Yuan model. The new model estimates 37% and 27% respectively for both routes.

Both models attribute the most important naphthalene production pathway to the reaction between 1-methyleneindene ($C_9H_6CH_2$) and H radical with more than 31% contribution while the major consumption pathway, as estimated by both models, is through OH attack on naphthalene to give 1-naphthalenyl radical and water. The updated model gives a better fit to the phenanthrene experimental data compared to the Yuan model, which we attribute to the improved description of C_7H_5 chemistry (R10 – R19

and R43 – R46). The most important pathway, based on ROP studies, for phenanthrene formation is through fulvenallenyl (C_7H_5) recombination, which is estimated to generate about 23% of phenanthrene by our model compared to $\sim 34\%$ predicted by the Yuan model while the most important consumption pathway is the formation of 4-phenanthrenyl radical and H_2 via H atom attack. This channel is predicted to contribute about 23% to phenanthrene depletion in both models.

We now examine the predicted concentration of C_5H_x species in the toluene flame (Fig. 7). Note that for these compounds, other than $c-C_5H_5$, isomer-resolved experimental measurements are not available, and these species (except $i-C_5H_3$, $c-C_5H_4$, and CH_2CCCCH_2) are not included in the Yuan model, so our analysis is based on the current model predictions alone. The mole fractions of C_5H_3 isomers, as predicted by the updated model, are shown in Fig. 7(a). Experimental and computational works of Yang et al. [6] and Hansen et al. [8]. Identified both $n-C_5H_3$ ($HCCCCHCH$) and $i-C_5H_3$ ($H_2CCCCCH$) to be the most abundant C_5H_3 isomers present in flames, making it imperative to include both species in DCKMs. The present study predicts $i-C_5H_3$ to be the dominant isomer in this toluene flame, by around a factor of 7. The ROP analysis indicates that reactions $R46$ ($C_7H_5 (+M) \leftrightarrow i-C_5H_3 + C_2H_2 (+M)$), $C_4H_2 + CH_2 \leftrightarrow i-C_5H_3 + H$ and $i-C_5H_3 + H \leftrightarrow CH_2CCCCH_2$ (C_5H_4) are the dominant pathways to $i-C_5H_3$ formation with 45%, 25% and 14% contributions respectively with 83% of the formed $i-C_5H_3$ depleted through $R39$ ($i-C_5H_3 + H \leftrightarrow CHCCHCCH_2$ (ethynylallene)) to form ethynylallene ($CH_2CCHCCH$) and 8% going to the formation of 1,3-pentadiyne (CH_3CCCCH) through equation $R37$ ($i-C_5H_3 + H \leftrightarrow C_5H_4-13$). Reactions $R8$ ($n-C_5H_3 + H \leftrightarrow CHCCH_2CCH$) and $R44$ ($C_7H_5 (+M) \leftrightarrow n-C_5H_3 + C_2H_2 (+M)$) contribute about 81% and 17%, respectively, to the production of $n-C_5H_3$ while its only consumption pathway is via $R9$ ($n-C_5H_3 + H \leftrightarrow CHCCHCCH_2$).

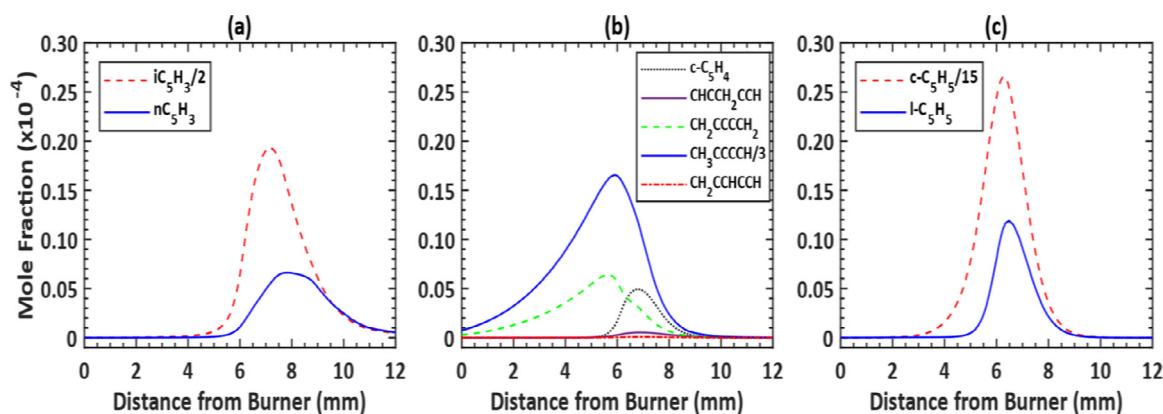


Fig. 7. Predicted mole fractions of key C_5H_x species, calculated with the updated chemical kinetic model. (a) C_5H_3 isomers, (b) C_5H_4 isomers, and (c) C_5H_5 isomers.

Figure 7(b) delineates the mole fraction profiles of C_5H_4 isomers. Aside from $c\text{-}C_5H_4$ and CH_2CCCCH_2 in the base model, three additional isomers were added to make up the C_5H_4 chemistry in the updated model. The different isomeric structures of C_5H_4 have quite distinct reaction pathways and therefore will have different contributions in combustion chemistry. 1,3-pentadiyne ($CH_3CCCCCH$) and penta-tetraene (CH_2CCCCH_2) appear to be the most dominant C_5H_4 species from ~ 0 –6 mm from the burner surface after which there is an onset of cyclopentadienylidene and 1,4-pentadiyne isomers. The onset of $c\text{-}C_5H_4$ and $CHCCH_2CCH$ coincides with the decline phase of other species which could imply a partial conversion of the most stable isomers, through a series of reactions, into the least stable [8] C_5H_4 forms ($c\text{-}C_5H_4$ and $CHCCH_2CCH$) at high temperatures. Reaction R37 ($i\text{-}C_5H_3 + H \leftrightarrow C_5H_4 + 13$) which favors the forward reaction (1,3-pentadiyne formation) is the only reaction featuring 1,3-pentadiyne ($CH_3CCCCCH$). This implies that its chemistry is still not well defined due to the absence of kinetic data. Based on ROP calculations, 65% and 35% of ethynylallene ($CHCCHCCH_2$) is formed through R39 ($i\text{-}C_5H_3 + H \leftrightarrow CHCCHCCH_2$) and R8 ($n\text{-}C_5H_3 + H \leftrightarrow CHCCHCCH_2$) respectively. However, ethynylallene ($CHCCHCCH_2$) has only one depletion pathway available to it, via R4 ($CHCCHCCH_2 + H \leftrightarrow CHCCH_2CCH + H$) which translates into 100% of $CHCCH_2CCH$ produced. The major consumption (72%) channel of this species, $CHCCH_2CCH$, is through its decomposition to propargyl and acetylene (R5). $c\text{-}C_5H_4$ is predominantly generated through the decomposition of C_7H_6 to $c\text{-}C_5H_4$ and C_2H_2 . This reaction contributes about 65% to $c\text{-}C_5H_4$ formation while the oxidation of $c\text{-}C_5H_5$ by OH to give $c\text{-}C_5H_4$ and H_2O has a contribution of 30%.

Figure 7(c) compares the linear C_5H_5 isomer, $l\text{-}C_5H_5$ (1-vinylpropargyl radical) predictions with the cyclic $c\text{-}C_5H_5$, cyclopentadienyl. It can be seen that the incorporation of $l\text{-}C_5H_5$ chemistry into the DCKM results in the prediction of significant levels of this isomer, up to 5% of the concentration of ubiquitous $c\text{-}C_5H_5$. This is consistent with experimental evidence [6,20,50–52] that suggests that a considerable amount of the open-chain $l\text{-}C_5H_5$ is produced alongside the more stable cyclic $c\text{-}C_5H_5$ (cyclopentadienyl) in flames, in radical reactions, and in pyrolysis.

4. Conclusion

This work has presented the chemical mechanism that may lead to the fragmentation of C_5H_5 isomers, $l\text{-}C_5H_5$ and $c\text{-}C_5H_5$, into $C_5H_4 + H$ and $C_3H_3 + C_2H_2$ channels as revealed by the experimental work of Shapero et al. [16]. Computational kinetic studies showed that the $C_3H_3 + C_2H_2$ pathway is preferred compared to the C_5H_4 channel which becomes increasingly significant at relevant combustion temperatures. Rate coefficient expressions de-

termined in this computational study, for the important reactions of $C_5H_4 + H$, were included in a detailed chemical kinetic model to simulate the chemical composition of a toluene flame at an equivalence ratio of 1.75. Other C_5H_x chemistries and related reactions absent in previous models have been incorporated herein to achieve a rigorous description of these species. The mole fraction predictions obtained were compared to those from a base model and were found to give a better representation of the published experimental mole fractions profiles of $c\text{-}C_5H_5$, and $C_{14}H_{10}$ species. This work will allow for the further development of the reaction chemistry of these C_5H_x compounds and incorporation into combustion simulations.

Supporting information available

Detailed reaction mechanism, thermodynamic, transport data files of chemical species, in Chemkin format, and flow reactor model prediction figures are supplied.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Australian Research Council (ARC) through Future Fellowship project FT130101304.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.12.046.

References

- [1] D.A. Kaden, R.A. Hites, W.G. Thilly, Mutagenicity of soot and associated polycyclic aromatic hydrocarbons to salmonella typhimurium, *Cancer Res.* 39 (1979) 4152–4159.
- [2] P.P. Fu, F.A. Beland, S.K. Yang, Cyclopenta-polycyclic aromatic hydrocarbons: potential carcinogens and mutagens, *Carcinogenesis* 1 (1980) 725–727.
- [3] J.L. Durant, W.F. Busby, A.L. Lafleur, B.W. Penman, C.L. Crespi, Human cell mutagenicity of oxygenated, nitrated and unsubstituted polycyclic aromatic hydrocarbons associated with urban aerosols, *Mutat. Res. Genet. Toxicol.* 371 (1996) 123–157.
- [4] M.F. Denissenko, A. Pao, M. Tang, G.P. Pfeifer, Preferential formation of benzo(a)pyrene adducts at lung cancer mutational hotspots in P53, *Science* 274 (1996) 430–432.
- [5] V. Ramanathan, G. Carmichael, Global and regional climate changes due to black carbon, *Nat. Geosci.* 1 (2008) 221–227.

- [6] B. Yang, C. Huang, L. Wei, J. Wang, L. Sheng, Y. Zhang, F. Qi, W. Zheng, W.K. Li, Identification of isomeric C_5H_3 and C_5H_5 free radicals in flame with tunable synchrotron photoionization, *Chem. Phys. Lett.* 423 (2006) 321–326.
- [7] H. Richter, J.B. Howard, Formation of polycyclic aromatic hydrocarbons and their growth to soot—a review of chemical reaction pathways, *Prog. Energy Combust. Sci.* 26 (2000) 565–608.
- [8] N. Hansen, S.J. Klippenstein, J.A. Miller, J. Wang, T.A. Cool, M.E. Law, P.R. Westmoreland, T. Kasper, K. Kohse-Höinghaus, Identification of C_5H_4 isomers in fuel-rich flames by photoionization mass spectrometry and electronic structure calculations, *J. Phys. Chem. A* 110 (2006) 4376–4388.
- [9] N. Hansen, T. Kasper, S.J. Klippenstein, P.R. Westmoreland, M.E. Law, C.A. Taatjes, K. Kohse-Höinghaus, J. Wang, T.A. Cool, Initial steps of aromatic ring formation in a laminar premixed fuel-rich cyclopentene flame, *J. Phys. Chem. A* 111 (2007) 4081–4092.
- [10] A. Yokoyama, X. Zhao, E.J. Hints, R.E. Continetti, Y.T. Lee, Molecular beam studies of the photodissociation of benzene at 193 and 248nm, *J. Chem. Phys.* 92 (1990) 4222–4233.
- [11] C.J. Pope, J.A. Miller, Exploring old and new benzene formation pathways in low-pressure premixed flames of aliphatic fuels, *Proc. Combust. Inst.* 28 (2000) 1519–1527.
- [12] A.M. Mebel, S.H. Lin, X.M. Yang, Y.T. Lee, Theoretical study on the mechanism of the dissociation of benzene. The $C_5H_3 + CH_3$ product channel, *J. Phys. Chem. A* 101 (1997) 6781–6789.
- [13] Y. Guo, X. Gu, F. Zhang, A.M. Mebel, R.I. Kaiser, Unimolecular decomposition of chemically activated pentatetraene ($H_2CCCCCH_2$) intermediates: a crossed beams study of dicarbon molecule reactions with allene, *J. Phys. Chem. A* 110 (2006) 10699–10707.
- [14] F. Goulay, D.L. Osborn, C.A. Taatjes, P. Zou, G. Meloni, S.R. Leone, Direct detection of polyynes formation from the reaction of ethynyl radical (C_2H) with propyne ($CH_3C\equiv CH$) and allene ($CH_2=C=CH_2$), *Phys. Chem. Chem. Phys.* 9 (2007) 4291–4300.
- [15] T.R. Melton, A.M. Vincitore, S.M. Senkan, The effects of equivalence ratio on the formation of polycyclic aromatic hydrocarbons and soot in premixed methane flames, *Symp. (Int.) Combust.*, 27 (1998), pp. 1631–1637.
- [16] M. Shapero, I.A. Ramphal, D.M. Neumark, Photodissociation of the cyclopentadienyl radical at 248nm, *J. Phys. Chem. A* 122 (2018) 4265–4272.
- [17] K. Roy, C. Horn, P. Frank, V.G. Slutsky, T. Just, High-temperature investigations on the pyrolysis of cyclopentadiene, *Symp. (Int.) Combust.* (1998), pp. 329–336.
- [18] A. Burcat, M. Dvinyaninov, Detailed kinetics of cyclopentadiene decomposition studied in a shock tube, *Int. J. Chem. Kinet.* 29 (1997) 505–514.
- [19] R.D. Kern, Q. Zhang, J. Yao, B.S. Jursic, R.S. Tranter, M.A. Greybill, J.H. Kiefer, Pyrolysis of cyclopentadiene: rates for initial C–H bond fission and the decomposition of $c\text{-}C_5H_5$, *Symp. (Int.) Combust.* (1998), pp. 143–150.
- [20] J.D. Savee, T.M. Selby, O. Welz, C.A. Taatjes, D.L. Osborn, Time- and isomer-resolved measurements of sequential addition of acetylene to the propargyl radical, *J. Phys. Chem. Lett.* 6 (2015) 4153–4158.
- [21] F. Stahl, P.V.R. Schleyer, H.F. Bettinger, R.I. Kaiser, Y.T. Lee, H.F. Schaefer, Reaction of the ethynyl radical, C_2H , with methylacetylene, CH_3CCH , under single collision conditions: implications for astrochemistry, *J. Chem. Phys.* 114 (2001) 3476–3487.
- [22] V.D. Knyazev, I.R. Slagle, Kinetics of the reaction between propargyl radical and acetylene, *J. Phys. Chem. A* 106 (2002) 5613–5617.
- [23] L.V. Moskaleva, M.C. Lin, Unimolecular isomerization/decomposition of cyclopentadienyl and related bimolecular reverse process: ab initio MO/statistical theory study, *J. Comput. Chem.* 21 (2000) 415–425.
- [24] A. Jamal, A.M. Mebel, An ab initio/RRKM study of the reaction mechanism and product branching ratios of the reactions of ethynyl radical with 1, 2-butadiene, *Chem. Phys. Lett.* 518 (2011) 29–37.
- [25] G. da Silva, Mystery of 1-vinylpropargyl formation from acetylene addition to the propargyl radical: an open-and-shut case, *J. Phys. Chem. A* 121 (2017) 2086–2095.
- [26] W. Yuan, Y. Li, P. Dagaut, J. Yang, F. Qi, Investigation on the pyrolysis and oxidation of toluene over a wide range conditions. II. A comprehensive kinetic modeling study, *Combust. Flame* 162 (2015) 22–40.
- [27] W. Yuan, Y. Li, P. Dagaut, J. Yang, F. Qi, Investigation on the pyrolysis and oxidation of toluene over a wide range conditions. I. Flow reactor pyrolysis and jet stirred reactor oxidation, *Combust. Flame* 162 (2015) 3–21.
- [28] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, G. Scalmani, V. Barone, G.A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A.V. Marenich, J. Bloino, B.G. Janesko, R. Gomperts, B. Mennucci, H.P. Hratchian, J.V. Ortiz, A.F. Izmaylov, J.L. Sonnenberg, D. Williams-Young, F. Ding, F. Lipparini, F. Egidi, J. Goings, B. Peng, A. Petrone, T. Henderson, D. Ranasinghe, V.G. Zakrzewski, J. Gao, N. Rega, G. Zheng, W. Liang, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, K. Throssell, J.A. Montgomery, J.E.P. Jr., F. Ogliaro, M.J. Bearpark, J.J. Heyd, E.N. Brothers, K.N. Kudin, V.N. Staroverov, T.A. Keith, R. Kobayashi, J. Normand, K. Raghavachari, A.P. Rendell, J.C. Burant, S.S. Iyengar, J. Tomasi, M. Cossi, J.M. Millam, M. Klene, C. Adamo, R. Cammi, J.W. Ochterski, R.L. Martin, K. Morokuma, O. Farkas, J.B. Foresman, D.J. Fox, Gaussian 16 Revision B. 01, (2016).
- [29] G. da Silva, G3X-K theory: a composite theoretical method for thermochemical kinetics, *Chem. Phys. Lett.* 558 (2013) 109–113.
- [30] J.R. Barker, T.L. Nguyen, J.F. Stanton, C. Aieta, M. Ceotto, F. Gabas, T.J.D. Kumar, C.G.L. Li, L.L. Lohr, A. Maranzana, MultiWell-2017 software suite, (2017).
- [31] J.H. Kiefer, R.S. Tranter, H. Wang, A.F. Wagner, Thermodynamic functions for the cyclopentadienyl radical: the effect of Jahn-Teller distortion, *Int. J. Chem. Kinet.* (2001).
- [32] G. da Silva, J.A. Cole, J.W. Bozzelli, Kinetics of the cyclopentadienyl + acetylene, fulvenallene + H, and 1-ethynylcyclopentadiene + H reactions, *J. Phys. Chem. A* 114 (2010) 2275–2283.
- [33] T. Lenzer, K. Luther, K. Reihs, A.C. Symonds, Collisional energy transfer probabilities of highly excited molecules from kinetically controlled selective ionization (KCSI). II. The collisional relaxation of toluene: $P(E', E)$ and moments of energy transfer for energies up to 50 000cm⁻¹, *J. Chem. Phys.* 112 (2000) 4090–4110.
- [34] ANSYS Chemkin-Pro 19.0 Reaction Workbench., (2018).
- [35] T.H. Chung, L.L. Lee, K.E. Starling, Applications of kinetic gas theories and multiparameter correlation for prediction of dilute gas viscosity and thermal conductivity, *Ind. Eng. Chem. Fundam.* 23 (1984) 8–3.
- [36] S.J. Pinches, G. da Silva, On the separation of timescales in chemically activated reactions, *Int. J. Chem. Kinet.* 45 (2013) 387–396.
- [37] G. da Silva, A.J. Trevitt, M. Steinbauer, P. Hemberger, Pyrolysis of fulvenallene (C_7H_6) and fulvenallenyl (C_7H_5): theoretical kinetics and experimental product detection, *Chem. Phys. Lett.* 517 (2011) 144–148.
- [38] G. da Silva, J.W. Bozzelli, The C_7H_5 fulvenallenyl radical as a combustion intermediate: potential new pathways to two- and three-ring PAHs, *J. Phys. Chem. A* 113 (2009) 12045–12048.
- [39] A.N.S.Y.S. Chemkin-Pro 19.0 PREMIX Application, (2018).
- [40] Y. Li, J. Cai, L. Zhang, T. Yuan, K. Zhang, F. Qi, Investigation on chemical structures of premixed toluene flames at low pressure, *Proc. Combust. Inst.* 33 (2011) 593–600.
- [41] J. Cai, W. Yuan, L. Ye, Z. Cheng, Y. Wang, L. Zhang, F. Zhang, Y. Li, F. Qi, Experimental and kinetic modeling study of 2-butanol pyrolysis and combustion, *Combust. Flame* 160 (2013) 1939–1957.
- [42] L.B. Harding, S.J. Klippenstein, Y. Georgievskii, On the combination reactions of hydrogen atoms with resonance-stabilized hydrocarbon radicals, *J. Phys. Chem. A* 111 (2007) 3789–3801.
- [43] G. da Silva, A.J. Trevitt, Chemically activated reactions on the C_7H_5 energy surface: propargyl + diacetylene, $i\text{-}C_5H_3$ + acetylene, and $n\text{-}C_5H_3$ + acetylene, *Phys. Chem. Chem. Phys.* 13 (2011) 8940–8952.
- [44] H. Jin, L. Xing, J. Hao, J. Yang, Y. Zhang, C.C. Cao, Y. Pan, A. Farooq, A chemical kinetic modeling study of indene pyrolysis, *Combust. Flame* 206 (2019) 1–20.
- [45] L. Pratali Maffei, M. Pelucchi, T. Faravelli, C. Cavallotti, Theoretical study of sensitive reactions in phenol decomposition, *React. Chem. Eng.* 5 (2020) 452–472.
- [46] G.R. Galimova, V.N. Azyazov, A.M. Mebel, Reaction mechanism, rate constants, and product yields for the oxidation of Cyclopentadienyl and embedded five-member ring radicals with hydroxyl, *Combust. Flame* 187 (2018) 147–164.
- [47] A.D. Oleinikov, V.N. Azyazov, A.M. Mebel, Oxidation of cyclopentadienyl radical with molecular oxygen: a theoretical study, *Combust. Flame* 191 (2018) 309–319.
- [48] H. Richter, S. Granata, W.H. Green, J.B. Howard, Detailed modeling of PAH and soot formation in a laminar premixed benzene/oxygen/argon low-pressure flame, (2020).
- [49] A.R. Ghildina, A.D. Oleinikov, V.N. Azyazov, A.M. Mebel, Reaction mechanism, rate constants, and product yields for unimolecular and H-assisted decomposition of 2,4-cyclopentadienone and oxidation of cyclopentadienyl with atomic oxygen, *Combust. Flame* 183 (2017) 181–193.
- [50] C. Huang, L. Wei, B. Yang, J. Wang, Y. Li, L. Sheng, Y. Zhang, F. Qi, Lean premixed gasoline/oxygen flame studied with tunable synchrotron vacuum UV photoionization, *Energy Fuels* 20 (2006) 1505–1513.
- [51] Y. Li, L. Zhang, Z. Tian, T. Yuan, J. Wang, B. Yang, F. Qi, Experimental study of a fuel-rich premixed toluene flame at low pressure, *Energy Fuels* 23 (2009) 1473–1485.
- [52] P. Hemberger, G. da Silva, A.J. Trevitt, T. Gerber, A. Bodi, Are the three hydroxyphenyl radical isomers created equal?—The role of the phenoxy radical, *Phys. Chem. Chem. Phys.* 17 (2015) 30076–30083.